SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.1: Cell Structure — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.1: Cell Structure
Driving Question	DRIVING QUESTION: "Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Healing Cut: Launching Our Investigation Into Why Cells Look Different**	Students examined a short narrative/video of a cut on the hand healing over several days — a context familiar to Kenyan learners from everyday farm or kitchen accidents. They noted visible changes: bleeding stops, a scab forms, new pink skin appears. Teacher should prompt: 'What do you think is happening INSIDE the skin that we cannot see?' Allow 3–4 cold-calls before revealing that scientists have discovered many different types of tiny structures working simultaneously beneath the surface.	Students began to appreciate that wound healing is not a single simple event — it is a coordinated series of biological processes. They were introduced to the anchoring phenomenon: the healing cut. They also generated their first Driving Question Board (DQB) entries, recording 'wonder questions' such as 'Where does the new skin come from?' and 'What is inside our skin?' Pedagogical note: Do NOT answer DQB questions yet — let curiosity build. Record all student questions visibly on the class DQB for revisiting throughout the unit.	At this early stage students cannot yet fully explain the phenomenon — this is intentional. Sample initial student explanation: 'The wound heals because the body fixes itself somehow.' Teacher should document these naive explanations (take a photo or note key phrases) so students can contrast them with their sophisticated final explanations in Lesson 12. This lesson establishes the epistemic need for the entire unit: we must understand cells before we can explain healing.
Lesson 2 **What Is a Cell? Introduction to Cell Theory**	Students examined historical timeline cards showing Hooke (cork cells, 1665), Leeuwenhoek (bacteria in pond water, 1670s), Schleiden (plant cells, 1838), Schwann (animal cells, 1839), and Virchow (cells from cells, 1855). Teacher should ask: 'Why did it take nearly 200 years to develop cell theory? What was limiting scientists?' (Expected answer: lack of good microscopes.) Connect to Kenya: just as maize farming knowledge built up over generations in the Rift Valley, scientific	Students learned the three tenets of cell theory verbatim: (1) All living things are made of one or more cells. (2) The cell is the basic structural and functional unit of life. (3) All cells arise from pre-existing cells. They applied these tenets to the anchoring phenomenon: new skin cells during wound healing must come from pre-existing skin cells (Virchow's tenet). This is the first direct conceptual link to healing. Pedagogical note: Require students to write all three tenets in their own words AND in scientific	Students updated their initial healing explanation: 'The wound heals because new cells are made from existing cells, and these cells fill the gap.' Teacher should probe: 'But what KIND of cells fill the gap? Are all cells the same?' — deliberately leaving this unanswered to drive curiosity into Lessons 3–9. Students add their first entry to the Summary Table and revise their model to include 'cells' at the wound site for the first time.

	understanding builds incrementally on prior evidence.	language — both versions matter for KICD assessment.	
Lesson 3 **Prokaryotic vs. Eukaryotic Cells: Two Worlds Inside One Body**	Students compared labelled electron micrograph diagrams of a prokaryotic cell (E. coli bacterium) and a eukaryotic cell (human white blood cell). Teacher directs: 'List THREE things you see in the eukaryotic cell that you do NOT see in the prokaryotic cell. You have 90 seconds — go.' Cold-call 4 students. Key observations: membrane-bound nucleus, mitochondria, endoplasmic reticulum. Connect to Kenya: bacteria like E. coli are relevant to food safety — githeri or nyama choma left unrefrigerated can harbour dangerous prokaryotes.	Students learned that prokaryotic cells (bacteria) lack a membrane-bound nucleus — their DNA floats freely in the nucleoid region — and have no membrane-bound organelles. Eukaryotic cells (all plant and animal cells) possess a membrane-bound nucleus and specialised membrane-bound organelles. Students also learned that the human body contains both: our own eukaryotic cells AND trillions of prokaryotic bacterial cells (microbiome). Wound healing relevance: bacteria that enter a cut are prokaryotes — the immune system's eukaryotic white blood cells engulf and destroy them.	Students refined their healing explanation: 'When a cut happens, bacteria (prokaryotic cells) enter the wound. The body's white blood cells (eukaryotic cells) recognise and destroy the bacteria. This is why the area becomes swollen — the immune response.' Pedagogical note: Many students conflate 'bacteria' with 'germs that are always harmful.' Address this misconception: our gut bacteria (e.g., in the digestive system that processes ugali and beans) are essential to our health. Not all prokaryotes cause disease.
Lesson 4 **Plant Cell Ultrastructure — Every Organelle Has a Farming Job**	Students observed a detailed labelled diagram of a plant cell and were asked to match each of the nine key organelles to a 'farming job' analogy: e.g., the chloroplast = solar panel on a Kericho tea farm (captures energy); the vacuole = water storage tank during dry season; the cell wall = fence around a shamba; the mitochondria = generator for night farming; the nucleus = farm manager's office. Teacher should allow pair discussion (2 minutes) before cold-calling 4–5 pairs.	Students learned the nine key plant cell organelles and their functions: cell wall (structural support/protection), cell membrane (selective barrier), nucleus (controls cell activities/contains DNA), chloroplasts (photosynthesis — light energy to glucose), large central vacuole (maintains turgor pressure, stores water/nutrients), mitochondria (cellular respiration — releases energy as ATP), endoplasmic reticulum (transport network — rough ER has ribosomes for protein synthesis), ribosomes (protein synthesis), and Golgi apparatus (packages and ships proteins). Wound healing relevance: This lesson sets the foundation for understanding that organelles are not random — structure always matches function.	Students added plant cell organelles to their science notebooks with function annotations. Teacher should note: plant cell ultrastructure is not directly relevant to the healing cut (skin is made of animal cells), but it is essential comparative knowledge that students will use in Lesson 5 and Lesson 9. Prompt students: 'When we get to skin cells, which of these organelles will we keep? Which will disappear? Why?' Record predictions on a sticky note inside notebooks for revisiting.
Lesson 5 **Animal Cell Ultrastructure — Comparing Plant and Animal Cells**	Students used a Venn diagram to compare plant and animal cells, sorting organelles into three zones: plant only / both / animal only. Teacher provides the organelle list; students sort independently (3 minutes), then compare with a partner, then class cold-call. Key finding: animal cells have lysosomes (absent in plant cells) and centrioles (absent in plant cells); plant cells	Students learned that animal cells contain lysosomes — membrane-bound sacs of digestive enzymes that break down waste materials, worn-out organelles, and pathogens. They also contain centrioles, which organise the spindle fibres during cell division. Critically for the anchoring phenomenon: lysosomes in white blood cells digest engulfed bacteria at the wound	Students significantly upgraded their healing explanation: 'Skin is made of animal cells. When a cut occurs, skin cells divide (using centrioles) to replace lost cells. White blood cells use lysosomes to destroy invading bacteria. The cell membrane controls what enters and leaves each cell during repair.' Teacher should cold-call 3 students to verbalise this explanation before

	have cell wall, chloroplasts, and large central vacuole (absent in animal cells). Both share: nucleus, cell membrane, mitochondria, ribosomes, ER, Golgi apparatus.	site; centrioles enable skin cells (keratinocytes) to divide and replace damaged tissue. This lesson is the most directly relevant to the healing cut so far — make this connection explicit.	students write it. Pedagogical note: Watch for the common misconception that lysosomes 'eat the whole cell' — clarify that lysosomes digest specific targets, not the host cell itself (unless during programmed cell death).
Lesson 6 **Microscopy Practical 1 — Observing Onion Epidermal Cells**	Students prepared wet mount slides of onion epidermal cells (peeled from a fresh red or white onion — readily available at any Kenyan market), stained with iodine solution, and observed under light microscopes at 40×, 100×, and 400× magnification. Teacher instruction: 'Draw what you see at EACH magnification. Label: cell wall, vacuole, nucleus. Note what you do NOT see and explain why.' Wait time: allow 8–10 minutes of silent observation and drawing before discussion. Common issue: students over-press the coverslip and burst cells — remind them to lower it at an angle.	Students observed that onion epidermal cells clearly show a cell wall (giving the brick-like rectangular shape), a large vacuole, and a nucleus — but NO chloroplasts. The critical reasoning task: onion bulbs grow underground and never receive sunlight, so chloroplasts (needed for photosynthesis) are not required. This reinforces the core principle: structure matches function. Students also practised microscopy skills — focusing technique, magnification calculation (eyepiece × objective), and biological drawing conventions (pencil, no shading, label lines with ruler).	Students connected this practical to the anchoring phenomenon by reasoning: 'Skin cells also lack chloroplasts because skin does not photosynthesise. Just as the onion cell's structure matches its underground storage function, skin cells will have structures that match their protective and repair function.' Pedagogical note: Biological drawing is a KICD-assessed skill — insist on correct conventions (sharp pencil, no shading, ruled label lines, magnification stated). Mark drawings formatively during the practical, not after.
Lesson 7 **Microscopy Practical 2 — Human Cheek Cells: Observing Animal Cell Structure**	Students gently scraped the inside of their cheek with a clean cotton swab, smeared the sample onto a glass slide, stained with methylene blue, and observed under the microscope. Teacher should model the technique first, emphasising sterile precautions (dispose of swabs safely, do not share). Students observed at 100× and 400×. Key observation: cells appear irregular/oval (no rigid cell wall), nucleus is clearly visible and stained dark blue, no chloroplasts, no large vacuole. Students also noted variability in cell size and shape between samples.	Students confirmed through direct observation that animal cells (human cheek epithelial cells) contain a cell membrane, cytoplasm, and a membrane-bound nucleus, but lack a cell wall, chloroplasts, and large central vacuole — exactly as predicted from Lesson 5. The irregular shape of cheek cells (compared to the rectangular onion cells) is directly caused by the absence of a rigid cell wall. This is the first time students SEE the structure-determines-shape principle with their own cells. Wound healing link: epithelial cells like cheek cells are the same category as skin (epidermal) cells — the cells that must regenerate after a cut.	Students made the most personal connection yet to the anchoring phenomenon: 'The cells I just looked at are the same TYPE as the cells in my skin. When I get a cut, cells like these — with a flexible membrane, no cell wall, and a nucleus containing my DNA — divide and migrate to close the wound.' Teacher should prompt: 'What organelle controls cell division in these cells?' (centrioles) 'What organelle destroys pathogens in the white blood cells nearby?' (lysosomes) — reviewing Lesson 5 in context. Update DQB: tick off 'How can we see cells?' and 'What do skin cells look like?'
Lesson 8 **Cell Specialisation: Meaning, Significance, and Specialised Animal Cells**	Students examined a gallery of six specialised animal cell diagrams/images: red blood cell (biconcave, no nucleus), nerve cell (long axon, dendrites), muscle cell (long, striated, multi-nucleate), white blood cell (irregular, flexible), sperm cell (flagellum, many mitochondria), and ciliated epithelial cell (hair-like cilia). Teacher task:	Students learned that cell specialisation is the process by which cells develop distinct structures that equip them for specific functions. The principle is: structure determines function. Key examples relevant to wound healing: (1) Red blood cells — biconcave disc, no nucleus, packed with haemoglobin → maximises surface area for	Students now produced their most detailed healing explanation to date: 'When I get a cut: (1) Platelets (specialised cell fragments) rush to the site and form a clot. (2) White blood cells (specialised with large lysosomes and flexible membranes) engulf bacteria. (3) Red blood cells (specialised to carry oxygen — biconcave, no nucleus)

	'For each cell, identify ONE structural feature that is unusual compared to a typical animal cell, and suggest what job that feature helps the cell do. You have 5 minutes.' Cold-call 6 students — one per cell.	oxygen transport to healing tissue. (2) White blood cells — irregular flexible shape, large lysosomes → can engulf pathogens at wound site. (3) Platelets (bonus) — tiny, no nucleus, sticky → clump to form a clot. Teacher should explicitly connect each cell to the anchoring phenomenon.	deliver oxygen to fuel the healing process. (4) Skin cells (keratinocytes) divide using centrioles to replace lost tissue.' This is a major milestone — students are now explaining the phenomenon using specific cell structures. Celebrate this growth explicitly.
Lesson 9 **Specialised Plant Cells: Form Meets Function in the Bean Seedling**	Students examined diagrams and/or prepared slides of five specialised plant cells from a bean seedling (common in Kenyan school gardens and farms): root hair cell (long extension, no chloroplasts), xylem vessel cell (hollow, lignified, no nucleus at maturity), phloem sieve tube cell (perforated end walls, companion cell), guard cell (kidney-shaped pair, chloroplasts), and mesophyll cell (loosely packed, many chloroplasts). Teacher connects to local agriculture: 'Farmers in Kisumu and the Rift Valley grow beans. Every cell in that seedling has a specific job. Let's find out why.'	Students learned that plant cells specialise just as animal cells do. Each of the five cell types has a specific structural adaptation: root hair cells have a long thin extension to increase surface area for water absorption (no chloroplasts — underground, no light); xylem vessels are hollow tubes with lignified walls — dead at maturity — forming a continuous pipe for water transport; phloem sieve tubes have perforated end plates for sugar transport; guard cells change shape (opening/closing stomata) due to changes in turgor pressure; mesophyll cells are packed with chloroplasts for maximum photosynthesis. Core principle reinforced: structure always matches function.	Students drew the parallel between plant and animal specialisation: 'Just as red blood cells are shaped for oxygen transport and white blood cells are shaped for defence, root hair cells are shaped for absorption and xylem cells are shaped for water transport. All cells — plant or animal — are specialised for their specific job.' Teacher should note: this lesson is not directly about wound healing, but it powerfully reinforces the universal principle of the driving question. Use the DQB to prompt: 'Does specialisation explain why cells look different?' Students should now confidently answer YES with multiple examples.
Lesson 10 **Cell Organisation: Cells → Tissues → Organs → Organ Systems → Organism**	Students arranged a set of cards showing the five levels of biological organisation — cell, tissue, organ, organ system, organism — using a human skin wound as the organising context. Card examples: keratinocyte (cell) → epidermis (tissue) → skin (organ) → integumentary system (organ system) → human body (organism). Teacher then extends: 'Apply the same hierarchy to a maize plant from a farm in Kitale. Start at the cell level.' Pair task (3 minutes) → cold-call 3 pairs. This dual application (animal + plant) reinforces that organisation is a universal biological principle.	Students learned that cells of the same type group together to form tissues; different tissues combine to form organs; organs working together form organ systems; organ systems combine to form a complete organism. For wound healing: skin (organ) is composed of epidermal tissue (tissue) made of keratinocytes (cells). The integumentary system (organ system) includes skin, hair, and nails. When a cut occurs, healing involves not just cells but the coordination of multiple tissues (connective tissue, epithelial tissue, nervous tissue for pain signals, vascular tissue for blood supply). This level of organisation is what makes complex healing possible.	Students wrote an organisational explanation of wound healing: 'A cut disrupts cells → tissues → the skin organ → and requires the integumentary, circulatory, and immune organ systems to work together to repair the damage.' Teacher pedagogical note: Many students treat the five levels as a simple memorisation list. Push deeper: 'Why does organisation matter? Could single cells heal a wound as effectively as organised tissues?' (No — organisation enables division of labour and coordination.) This distinction is essential for KICD higher-order assessment questions.
Lesson 11 **Synthesis and Model Revision: Putting the Cell Story Together**	Students spread out all their Summary Table entries from Lessons 1–10 and were given 10 minutes to silently re-read their own progression of observations, learning, and explanations. Teacher prompt: 'Find THREE places where your understanding	Students consolidated the full arc of learning: plant cells and animal cells share core organelles (nucleus, cell membrane, mitochondria, ribosomes, ER, Golgi) but differ in presence of cell wall, large central vacuole, and chloroplasts (plant only) and	Students produced their penultimate wound healing explanation — the most sophisticated yet — integrating: cell theory (new cells from existing cells), cell types (prokaryote vs. eukaryote), organelle function (lysosomes, centrioles,

	changed significantly. Put a star next to each.' Then: 'Find ONE thing you wrote early on that you now know was incomplete or wrong.' Cold-call 4–5 students to share — normalise scientific revision as a strength, not a weakness. Students then created a revised cumulative model of wound healing incorporating all lessons.	lysosomes and centrioles (animal only). All cells — regardless of specialisation — share the same DNA but express different genes to produce different structures. The five levels of organisation explain how specialised cells achieve coordinated healing. This lesson is pedagogically critical: it makes the implicit learning explicit. Do not skip the model revision activity — it is the most powerful consolidation tool in the unit.	mitochondria, nucleus), specialisation (red blood cells, white blood cells, platelets, keratinocytes), and levels of organisation (cell → tissue → organ → system → organism). Teacher should use this as a formative assessment — read 5–6 student explanations before Lesson 12 and identify remaining gaps or misconceptions to address in the final lesson.
Lesson 12 **Consolidation, DQB Closure, and Final Explanation — The Healing Cut Fully Explained**	Students returned to the original anchoring phenomenon from Lesson 1 — the healing cut narrative — and re-read it with fresh eyes. Teacher asks: 'Read this again. How many things can you NOW explain that you could NOT explain in Lesson 1?' Students annotate the text individually (5 minutes), then share in pairs. The class then formally closes the DQB together: teacher reads each original student question aloud, and students vote on whether it has been answered, partially answered, or remains open. All three outcomes are valid and celebrated.	Students consolidated that all specialised cells in the body originate from the same organism and carry the same DNA, but different genes are expressed in different cells, producing different structures and functions — this is the fundamental answer to the driving question. The complete wound healing explanation integrates every lesson: cell theory explains why new cells appear; prokaryote/eukaryote distinction explains bacteria vs. immune cells; organelle knowledge explains HOW each cell does its job; specialisation explains WHY different cells are present; levels of organisation explain HOW they coordinate. Remaining DQB questions (e.g., 'How do cells know when to divide?' 'What signals trigger healing?') are acknowledged as genuine open scientific questions — modelling authentic scientific practice.	Students wrote their FINAL complete explanation of the anchoring phenomenon. A strong Grade 10 response should include: (1) Cell theory — new skin cells arise from pre-existing skin cells by cell division. (2) Specialisation — platelets clot the blood; white blood cells (with lysosomes) engulf bacteria; red blood cells (biconcave, no nucleus) deliver oxygen; keratinocytes (skin cells) migrate and divide to close the wound. (3) Organelle function — centrioles organise cell division; mitochondria provide ATP energy; lysosomes digest pathogens. (4) Levels of organisation — healing occurs at cell, tissue, organ, and organ system levels simultaneously. Teacher should compare each student's final explanation to their Lesson 1 prediction and make this growth visible. Celebrate the journey from 'wounds just glue back together' to a full cell-biology explanation.